



Session 12A1: Complex Numbers — The Matrix You Already Know

Phase 2 — Classical Techniques | 80 min

Prerequisite: Session 12A2 (Matrices and Vectors) — you know what a rotation matrix is.

The One Insight That Changes Everything

You already know that $R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ rotates vectors by θ .

Now look at this:

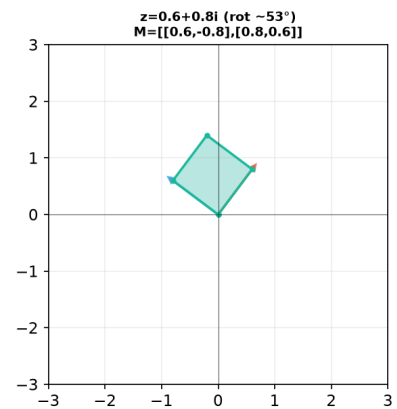
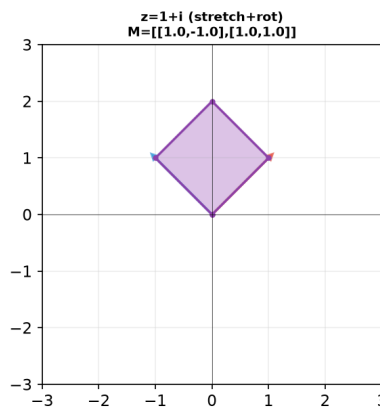
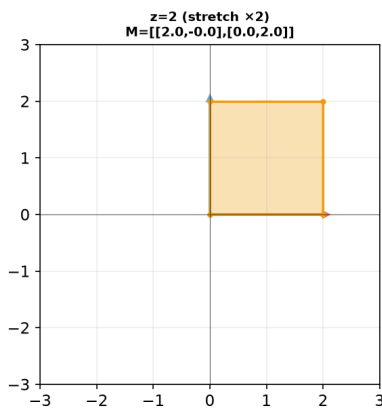
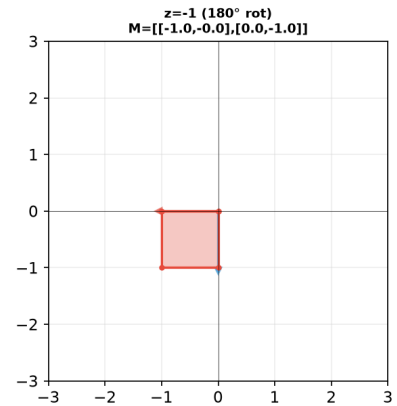
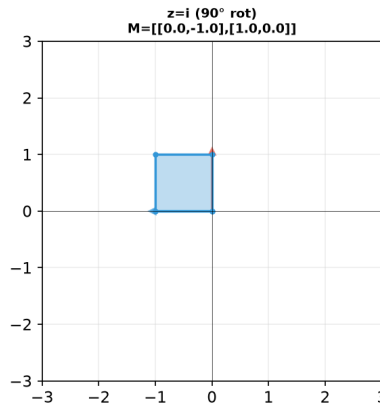
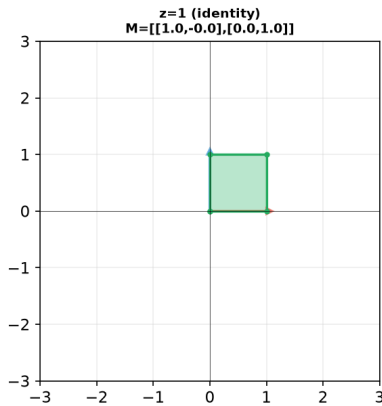
$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix} = \sqrt{a^2 + b^2} \cdot \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

This is a rotation-scaling matrix. It rotates by $\theta = \arctan(b/a)$ and scales by $r = \sqrt{a^2 + b^2}$.

A **complex number** $z = a + bi$ is just a shorthand for exactly this matrix.

Every operation you do with complex numbers — addition, multiplication, powers, roots — is an operation on these rotation-scaling matrices. You already know the geometry. Now we give it a name and a more compact notation.

Every Complex Number IS a Rotation-Scaling Matrix
 $a+bi \leftrightarrow \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$



Key: $a + bi \longleftrightarrow \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$. Addition is matrix addition. Multiplication is matrix multiplication. The modulus $|z| = \sqrt{a^2 + b^2}$ is the scaling factor. The argument $\arg(z) = \arctan(b/a)$ is the rotation angle.

Part A: The Basics — Arithmetic Through the Matrix Lens

Example 1: i — The 90° Rotation Matrix

$i = 0 + 1i$ corresponds to $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ — exactly the 90° CCW rotation matrix from 12A2.

This is why $i^2 = -1$: rotating 90° twice gives 180° , which is $-I = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$, which corresponds to -1 .

The 4-cycle — multiply by i repeatedly:

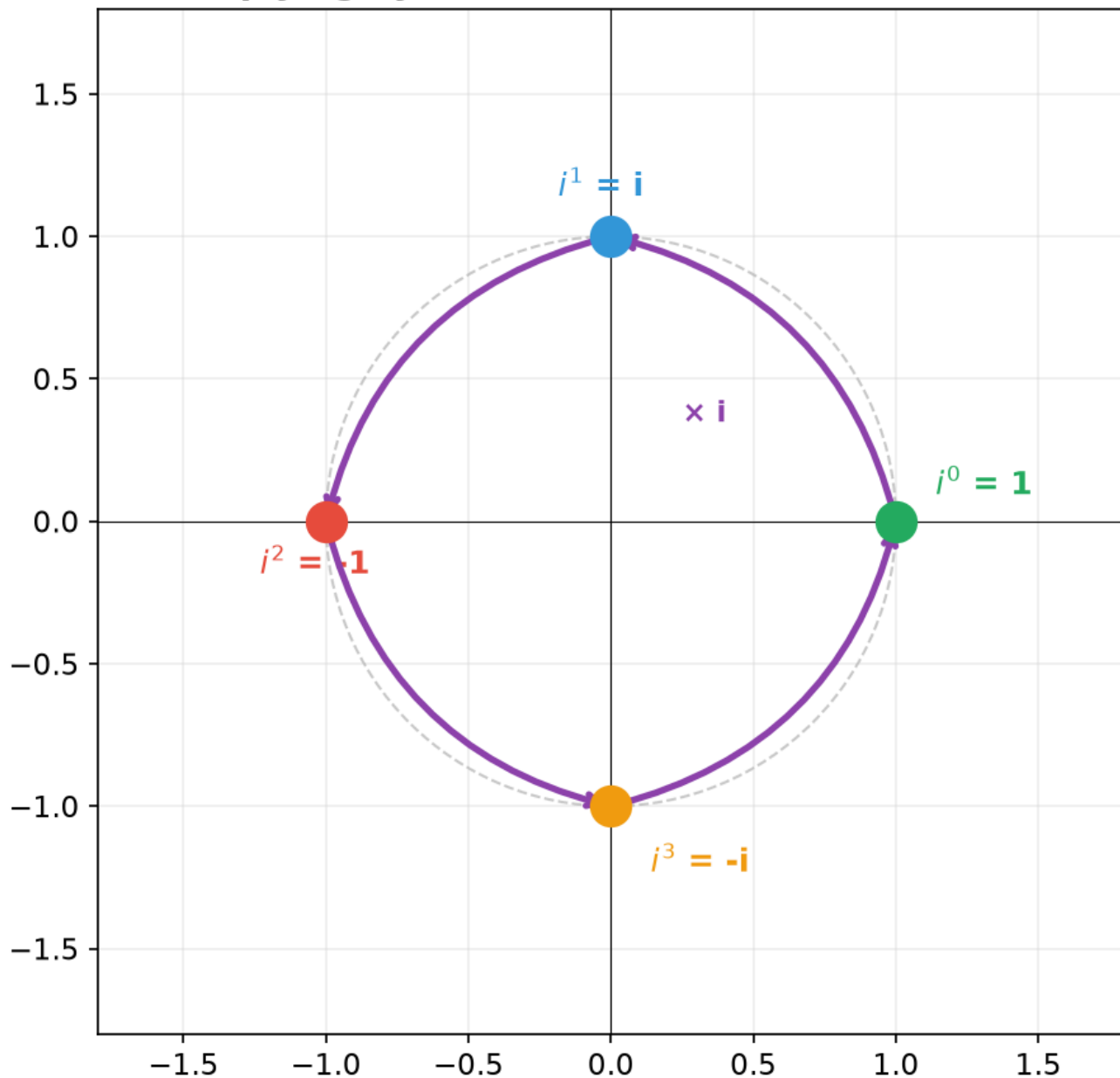
$i^1 = i$ (rotate 90°), $i^2 = -1$ (rotate 180°), $i^3 = -i$ (rotate 270°), $i^4 = 1$ (rotate 360°), $i^5 = i \dots$

To find i^{50} : $50 \div 4 = 12$ remainder 2. So $i^{50} = i^2 = -1$.

To find i^{101} : $101 \div 4 = 25$ remainder 1. So $i^{101} = i^1 = i$.

Square roots of negative numbers: $\sqrt{-9} = 3i$, $\sqrt{-8} = 2\sqrt{2}i$, $\sqrt{-1} = i$.

Multiplying by $i = 90^\circ$ Rotation on the Unit Circle



Matrix view: i^n is the rotation matrix R_{90° raised to the n th power. The 4-cycle is just the fact that $R_{90^\circ}^4 = I$.

Example 2: Adding and Multiplying — Same as Matrices

Let $z_1 = 2 + 3i$, $z_2 = 1 - 5i$.

Addition — exactly vector addition (component-wise, same as matrix addition):

$$(2 + 1) + (3 - 5)i = 3 - 2i.$$

Multiplication — exactly matrix multiplication (distribute, $i^2 \rightarrow -1$):

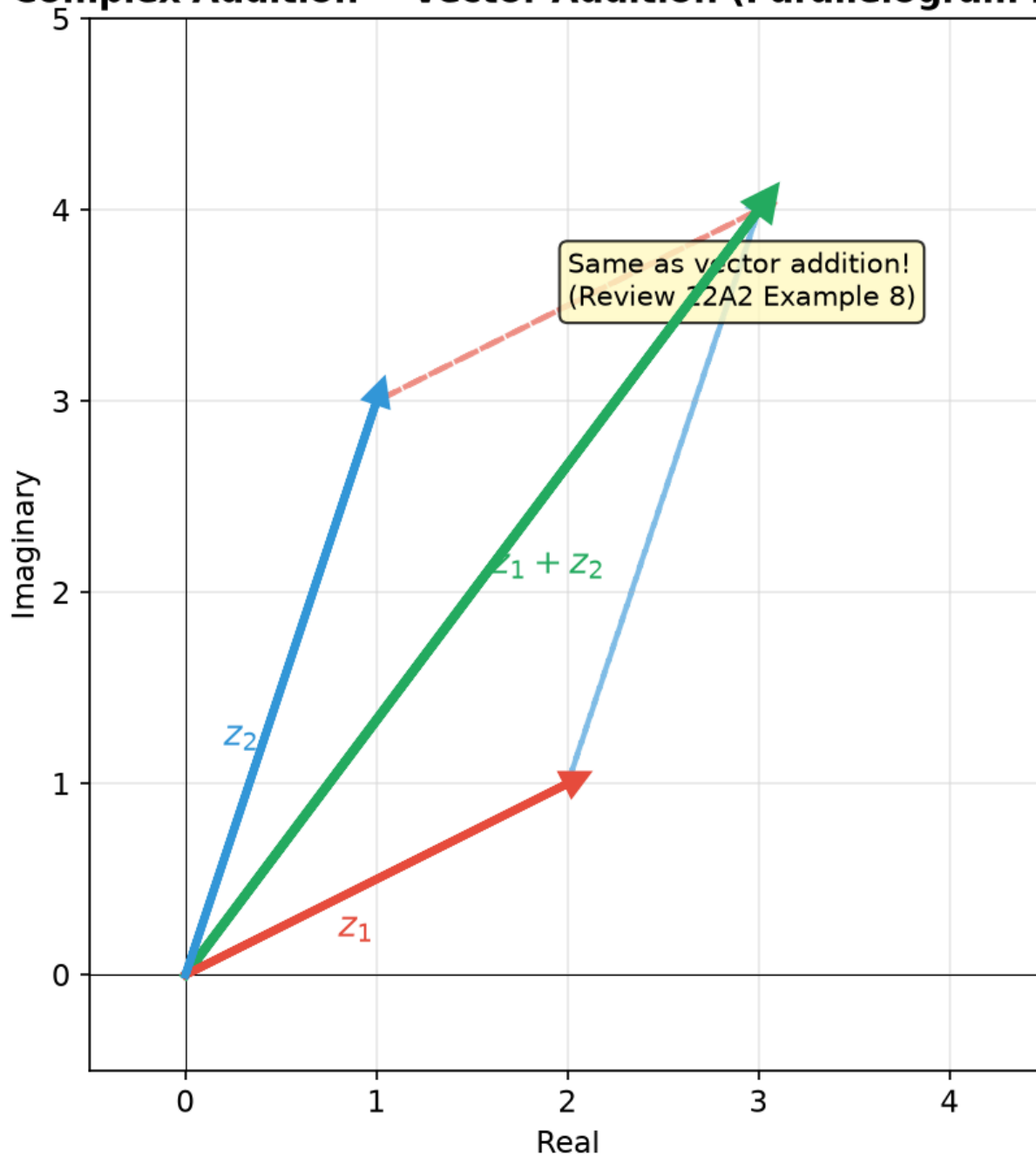
$$(2 + 3i)(1 - 5i) = 2 - 10i + 3i - 15i^2 = 2 - 7i + 15 = 17 - 7i.$$

Check via matrices: $\begin{pmatrix} 2 & -3 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} 1 & 5 \\ -5 & 1 \end{pmatrix} = \begin{pmatrix} 2 + 15 & 10 - 3 \\ 3 - 10 & 15 + 2 \end{pmatrix} = \begin{pmatrix} 17 & 7 \\ -7 & 17 \end{pmatrix}$, which corresponds to $17 - 7i$ ✓.

Division — multiply top and bottom by the conjugate:

$$\frac{2+3i}{1-5i} \cdot \frac{1+5i}{1+5i} = \frac{2+10i+3i+15i^2}{1+25} = \frac{2+13i-15}{26} = -\frac{1}{2} + \frac{1}{2}i.$$

Complex Addition = Vector Addition (Parallelogram Law)



Matrix view: Complex addition is matrix addition. Complex multiplication is matrix multiplication. There is nothing new — you already know these operations.

Example 3: Conjugate and Modulus — Reflection and Determinant

For $z = a + bi$, the matrix is $M = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$.

- **Conjugate:** $\bar{z} = a - bi$ corresponds to $M^T = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$ — the transpose.

Geometrically: reflection across the real axis.

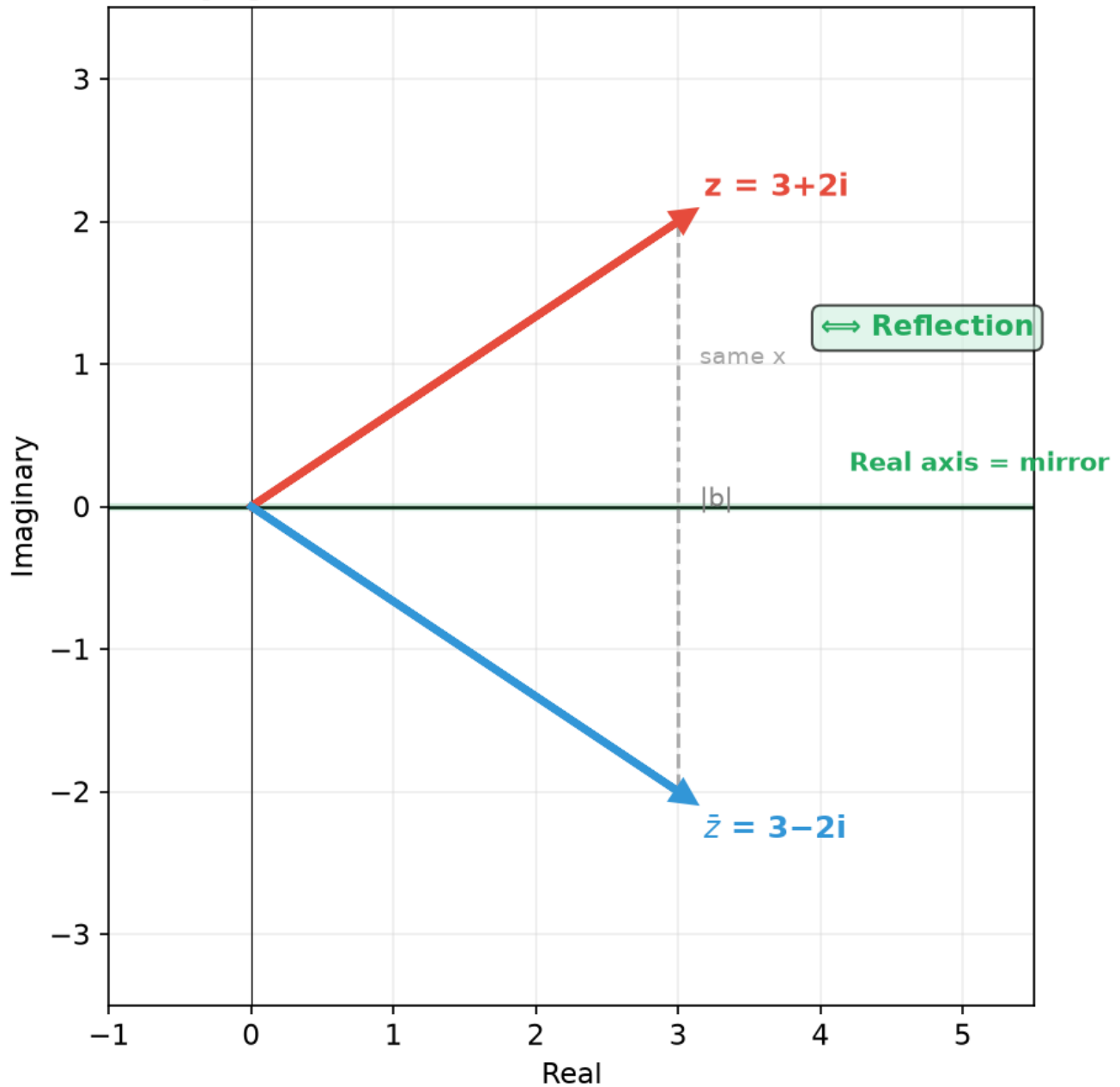
- **Modulus:** $|z| = \sqrt{a^2 + b^2}$. Notice: $\det(M) = a^2 + b^2 = |z|^2$.

Key identity: $z \cdot \bar{z} = a^2 + b^2 = |z|^2$. In matrix form: $M \cdot M^T = \det(M) \cdot I$.

For $z = 3 + 4i$: $\bar{z} = 3 - 4i$, $|z| = \sqrt{9 + 16} = 5$. Check: $(3 + 4i)(3 - 4i) = 25 = 5^2 \checkmark$.

Also: $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$. In matrices: $(AB)^T = B^T A^T$. Same property.

Conjugate = Reflection Across the Real Axis



Matrix view: The conjugate is the transpose. The modulus squared is the determinant. Everything connects.

Part B: The Complex Plane — Geometry You Already Know

Example 4: The Complex Plane and Polar Form

Plot $a + bi$ as the point (a, b) . This is exactly the same as plotting the vector (a, b) from 12A2.

The distance from the origin is $r = |z| = \sqrt{a^2 + b^2}$. The angle from the positive real axis is $\theta = \arg(z)$.

$$a = r \cos \theta, b = r \sin \theta.$$

Polar form: $z = r(\cos \theta + i \sin \theta) = re^{i\theta}$.

Example: $z = 1 + i\sqrt{3}$.

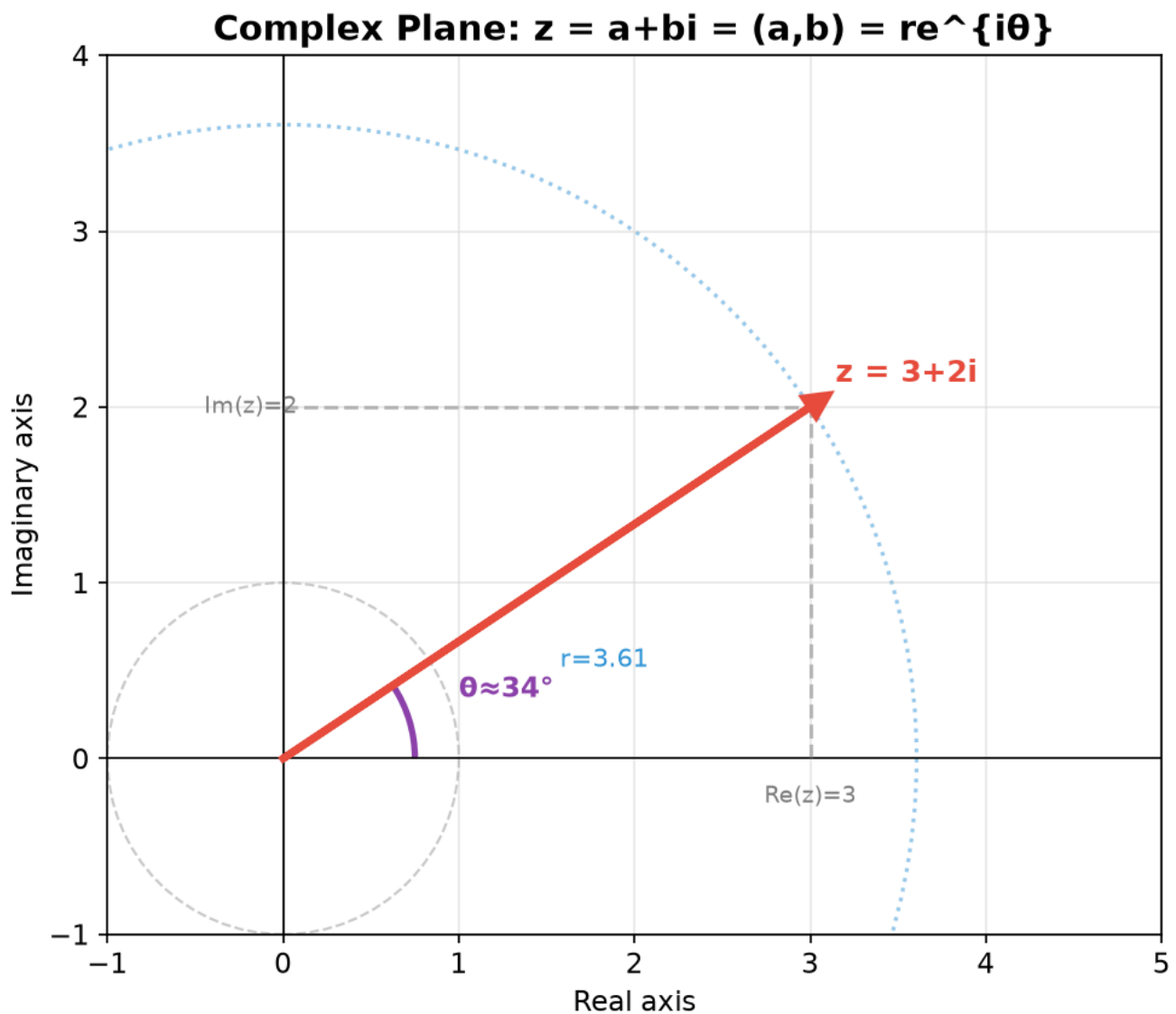
$$r = \sqrt{1 + 3} = 2. \theta = \arctan(\sqrt{3}/1) = \pi/3.$$

$$z = 2e^{i\pi/3}.$$

Example: $z = -1 + i$.

$$r = \sqrt{2}. \theta = 3\pi/4 \text{ (Quadrant II — check the signs!).}$$

$$z = \sqrt{2}e^{i\cdot 3\pi/4}.$$



Matrix view: $re^{i\theta}$ is exactly the rotation-scaling matrix with scaling factor r and rotation angle θ .

Example 5: Complex Multiplication = Stretch + Rotate

Multiplying $z_1 = r_1e^{i\theta_1}$ by $z_2 = r_2e^{i\theta_2}$:

$$z_1z_2 = (r_1e^{i\theta_1})(r_2e^{i\theta_2}) = (r_1r_2)e^{i(\theta_1+\theta_2)}$$

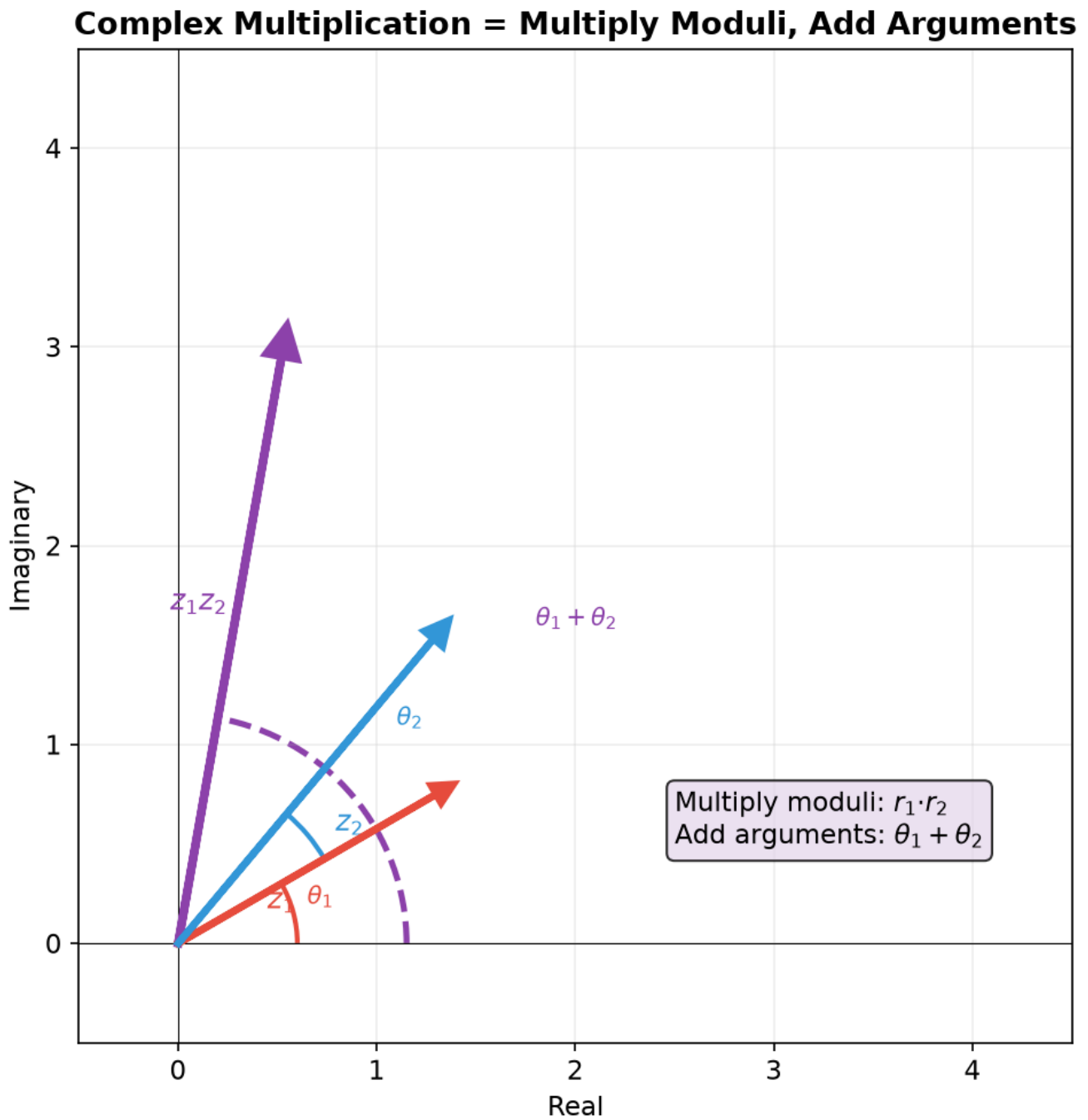
Moduli multiply: $r_1 \cdot r_2$. Arguments add: $\theta_1 + \theta_2$.

This is exactly matrix multiplication of two rotation-scaling matrices:

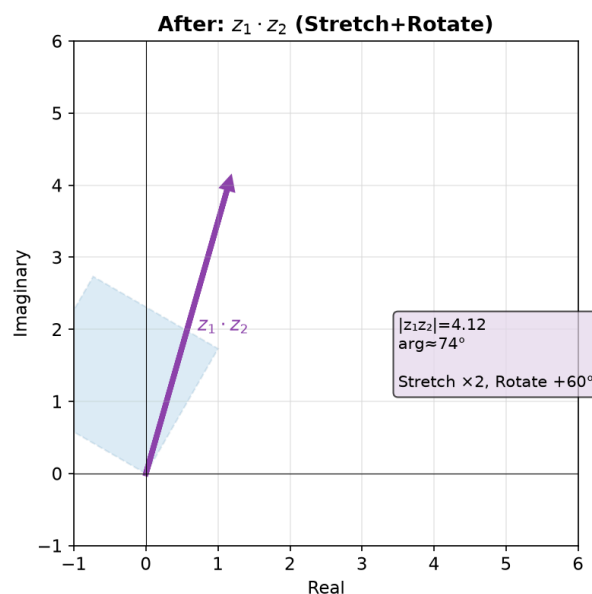
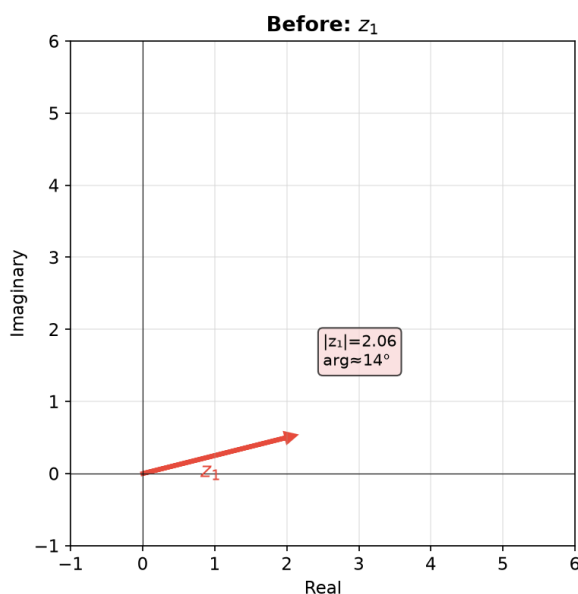
$$R_{\theta_1} S_{r_1} \cdot R_{\theta_2} S_{r_2} = R_{\theta_1 + \theta_2} S_{r_1 r_2}.$$

Example: Multiply $2e^{i\pi/6}$ by $3e^{i\pi/3}$:

Result: $6e^{i\pi/2} = 6i$. Stretch by 6, rotate to 90° .



Complex Multiplication = Rotation + Scaling
 $z = re^{i\theta}$ acts like rotation-scaling matrix $\begin{bmatrix} r \cos \theta & -r \sin \theta \\ r \sin \theta & r \cos \theta \end{bmatrix}$



Matrix view: This is why multiplying by i (which is $1 \cdot e^{i\pi/2}$) rotates by 90° without stretching — i is a pure rotation matrix.

Example 6: Euler's Formula and De Moivre — The Power Tools

Euler's formula: $e^{i\theta} = \cos \theta + i \sin \theta$.

Matrix form: $e^{i\theta} \longleftrightarrow R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$.

This compresses polar form to a single exponential: $z = re^{i\theta}$.

And the beautiful identity: $e^{i\pi} + 1 = 0$.

De Moivre's theorem: $z^n = r^n e^{in\theta}$.

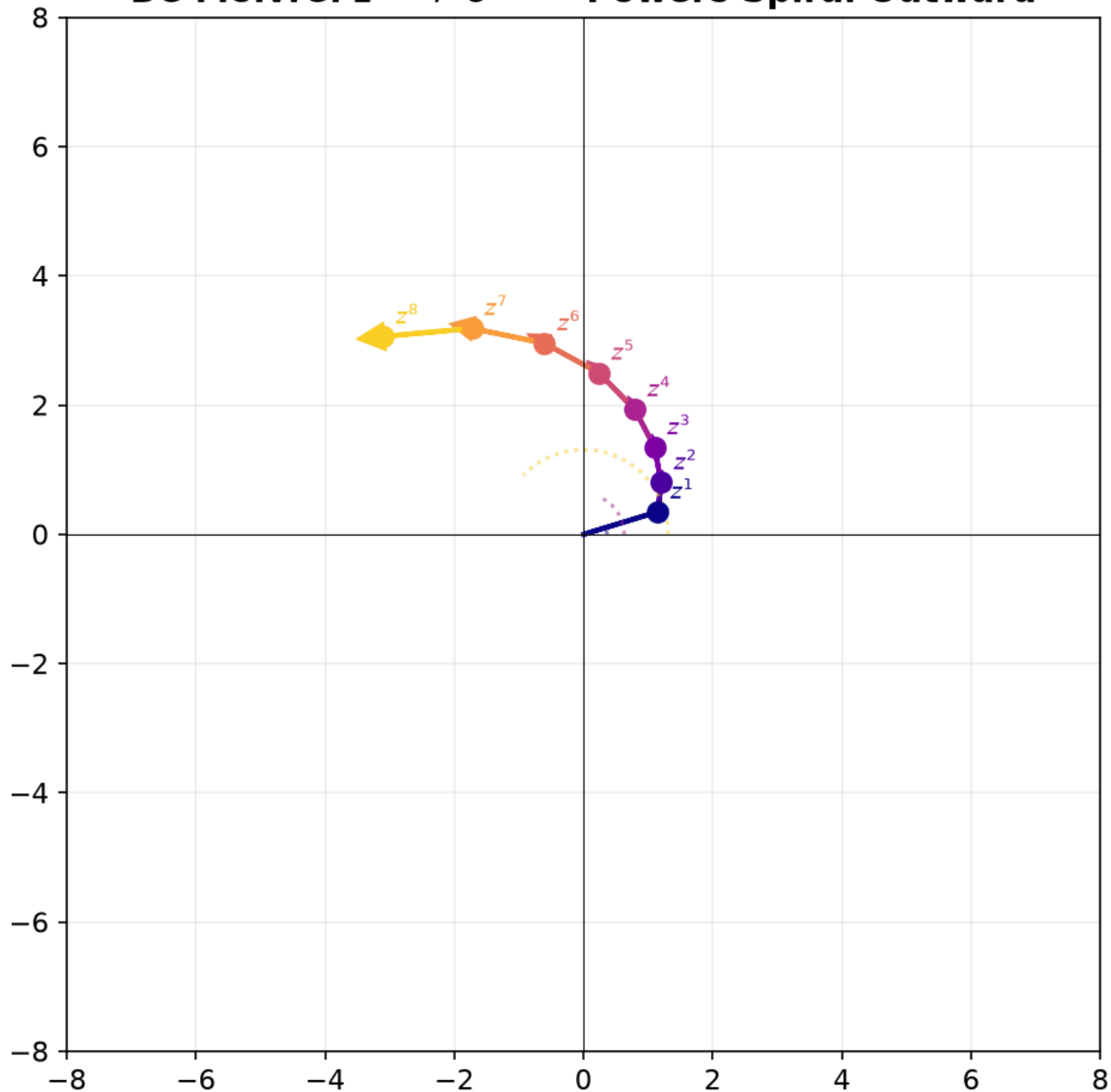
In matrix language: raising a rotation-scaling matrix to the n th power scales the factor by r^n and multiplies the angle by n . This is exactly what we did in 12A2 Example 7 (Matrix Powers).

Example: Compute $(1 + i)^6$.

$r = \sqrt{2}, \theta = \pi/4$.

$(1 + i)^6 = (\sqrt{2})^6 \cdot e^{i \cdot 6\pi/4} = 8 \cdot e^{i \cdot 3\pi/2} = 8(0 - i) = -8i$.

De Moivre: $z^n = r^n e^{in\theta}$ — Powers Spiral Outward



Matrix view: De Moivre is just matrix power M^n . Since M is a rotation-scaling matrix, M^n scales by r^n and rotates by $n\theta$. Nothing new.

Part C: Roots, Polynomials, and the Big Picture

Example 7: n th Roots of Unity — Regular Polygons

The equation $z^n = 1$ has exactly n solutions:

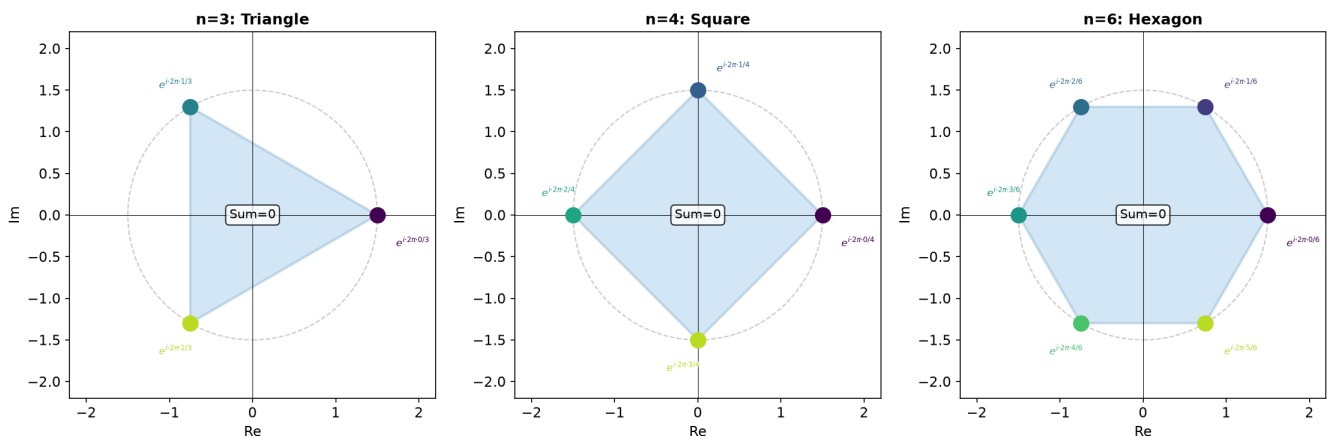
$$z_k = e^{i \cdot 2\pi k / n} \text{ for } k = 0, 1, \dots, n-1.$$

These are n points equally spaced on the unit circle — a **regular n -gon**. Their sum is always 0.

$n = 4$: $1, i, -1, -i$ — a square.

$n = 3$: $1, e^{i \cdot 2\pi/3}, e^{i \cdot 4\pi/3}$ — an equilateral triangle.

Roots of Unity = Regular n -gon on the Unit Circle



Example: Solve $z^3 = -8$.

$-8 = 8e^{i\pi}$. Three cube roots: $z_k = 2 \cdot e^{i(\pi+2\pi k)/3}$ for $k = 0, 1, 2$.

$$k = 0: 2e^{i\pi/3} = 1 + i\sqrt{3}.$$

$$k = 1: 2e^{i\pi} = -2.$$

$$k = 2: 2e^{i \cdot 5\pi/3} = 1 - i\sqrt{3}.$$

Three points equally spaced by 120° on a circle of radius 2.

Example 8: Reciprocal — Inversion in the Unit Circle

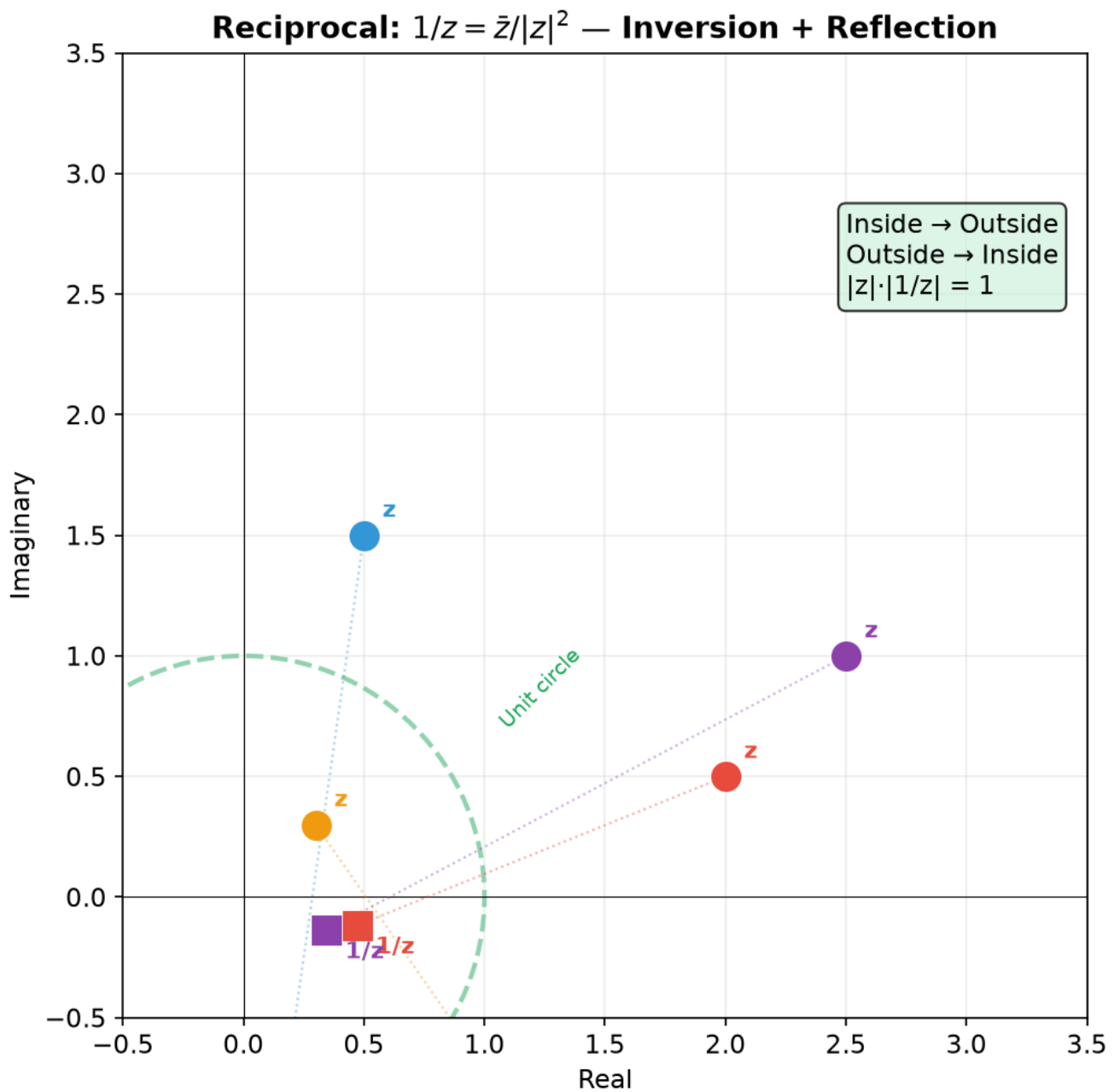
$$\frac{1}{z} = \frac{\bar{z}}{|z|^2} = \frac{1}{r}e^{-i\theta}.$$

Geometrically: invert the modulus ($r \rightarrow 1/r$) and reflect across the real axis ($\theta \rightarrow -\theta$).

- If $|z| > 1$ (outside), then $|1/z| < 1$ (inside).
- If $|z| < 1$ (inside), then $|1/z| > 1$ (outside).

The unit circle is the fixed boundary.

In matrix language: $1/z$ corresponds to $M^{-1} = \frac{1}{\det(M)} M^T$. The inverse of a rotation-scaling matrix divides by the determinant and transposes (reflects).



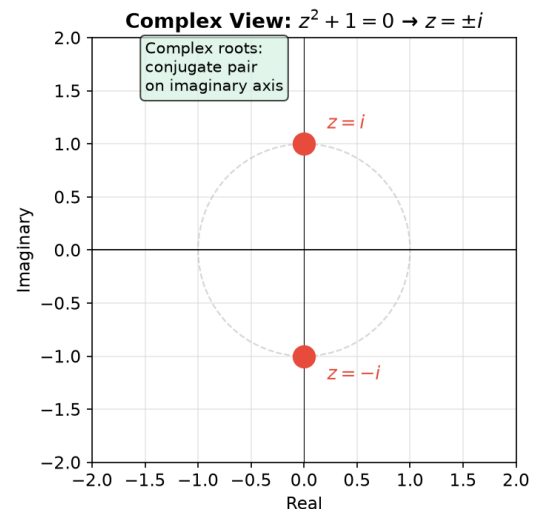
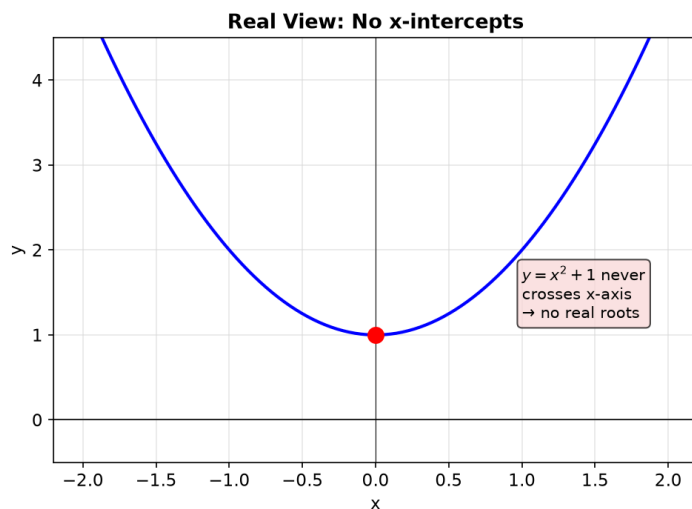
Example 9: Fundamental Theorem of Algebra — Why We Need Complex Numbers

Theorem: Every polynomial of degree n has exactly n complex roots (counting multiplicity).

Why this matters: Over the real numbers, $x^2 + 1 = 0$ has no solution. Over the complex numbers, it has two: $\pm i$. The complex numbers **algebraically close** the real numbers — you never need to invent new number systems to solve polynomial equations.

Complex roots come in conjugate pairs: If $a + bi$ is a root of a polynomial with real coefficients, then $a - bi$ is also a root.

Fundamental Theorem of Algebra: Degree $n \rightarrow$ Exactly n Complex Roots
 $x^2 + 1 = 0$ has two complex roots (conjugate pair)



Example: $z^2 + 2z + 5 = 0$.

$$z = \frac{-2 \pm \sqrt{4 - 20}}{2} = \frac{-2 \pm \sqrt{-16}}{2} = \frac{-2 \pm 4i}{2} = -1 \pm 2i.$$

The roots are a conjugate pair.

Up to here: Complex numbers are rotation-scaling matrices in compact notation. i is the 90° rotation. Polar form $re^{i\theta}$ = stretch by r , rotate by θ . De Moivre = matrix powers. Roots of unity = regular polygons. Every polynomial has all its roots in $\mathbb{C}$.

Visual Interlude: Five Geometric Views of Complex Numbers

View 1 — The Complex Plane as $\mathbb{R}^2$. $a + bi \leftrightarrow (a, b)$. Addition = vector addition. Conjugate = reflection across the real axis. Modulus = Euclidean distance.

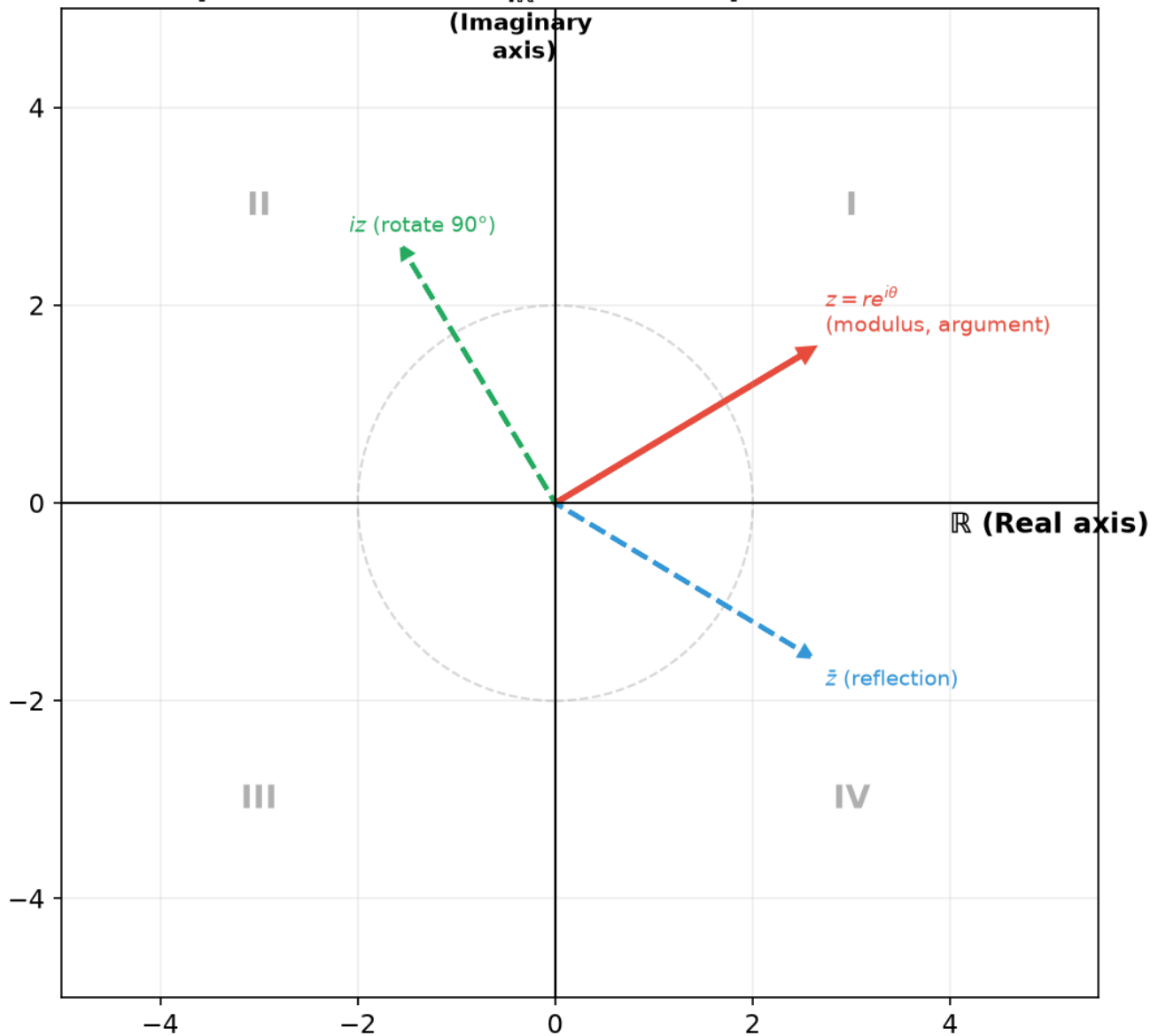
View 2 — Polar Form as Stretch + Rotate. $re^{i\theta}$: r is the scaling factor, θ is the rotation angle. Multiplication: multiply moduli, add arguments. Division: divide moduli, subtract arguments.

View 3 — Matrix Isomorphism. $a + bi \leftrightarrow \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$. $|z|^2 = \det(M)$. $\bar{z} \leftrightarrow M^T$. $1/z \leftrightarrow M^{-1}$.

View 4 — Roots as Regular Polygons. $z^n = w$ has n solutions, equally spaced on a circle of radius $|w|^{1/n}$. The sum of all n th roots of unity is 0.

View 5 — Algebraic Completeness. $\mathbb{C}$ is algebraically closed. Every polynomial factors completely over $\mathbb{C}$. This is why complex numbers exist — to guarantee that every polynomial equation has all its solutions.

The Complex Plane — All Geometric Operations at a Glance



Common Mistakes

Mistake 1: $\sqrt{-4} = -2$

Wrong path: "The square root of -4 is -2 ." $(-2)^2 = 4$, not -4 .

Right path: $\sqrt{-4} = \sqrt{4} \cdot \sqrt{-1} = 2i$.

Mistake 2: Forgetting the quadrant when computing $\arg(z)$

Wrong path: " $z = -1 - i$, so $\theta = \arctan(1) = 45^\circ$."

Why wrong: $\arctan(1) = 45^\circ$ but $(-1, -1)$ is in Quadrant III, where $\theta = 225^\circ$ (or -135°).

Right path: Always draw the point. Check which quadrant. Add π if in QII or QIII. Use $\arctan 2(b, a)$.

Mistake 3: Confusing $|z|^2$ with z^2

Wrong path: " $|3 + 4i|^2 = (3 + 4i)^2$." $|z|^2 = 25$, but $z^2 = -7 + 24i$.

Right path: $|z|^2 = z\bar{z} = \det(M)$ (always real and ≥ 0). z^2 is generally complex.

Mistake 4: Thinking i is the only "imaginary" thing

Wrong path: "Complex numbers are made-up, i is mysterious."

Why wrong: Complex numbers are exactly rotation-scaling matrices. i is the 90° rotation matrix.

Right path: $a + bi$ is compact notation for $\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$. Everything is grounded in matrices you already understand.

What We Just Did

- (1) Core insight – $a+bi$ IS the rotation-scaling matrix $\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$.
 $i = 90^\circ$ rotation. $i^2 = -1$ because two 90° rotations = $180^\circ = -I$.
- (2) Arithmetic – addition = vector addition. Multiplication = matrix mult.
Conjugate = transpose = reflection. $|z|^2 = \det(M)$. $1/z = M^{-1}$.
- (3) Polar form – $z = re^{i\theta}$: $r = |z| = \sqrt{\det(M)}$, $\theta =$ rotation angle.
Multiplication: multiply moduli, add angles (matrix composition).
De Moivre: $z^n = r^n e^{in\theta}$ (matrix power – same as 12A2 Example 7).
- (4) Roots – $z^n = w$ has n solutions (regular n -gon).
Fundamental Theorem: every degree- n polynomial has n complex roots.

Practice 1

Divide $\frac{3-i}{2+i}$. Write in $a + bi$ form. Then write the corresponding 2×2 matrix for the result and verify its determinant equals $|z|^2$.

→ Reference: **Example 2, 3**

Solutions: [Solutions](#)

Practice 2

Write $z = 1 - i$ in polar form. Compute z^8 using De Moivre. Then explain why the answer must be a positive real number using the matrix view (what happens to the rotation angle after 8 steps?).

→ Reference: **Example 6**

Solutions: [Solutions](#)

Practice 3

Find all three cube roots of -8 . Give each in $a + bi$ form. Write the rotation-scaling matrix for each root and verify that cubing the matrix gives $-8I$.

→ Reference: **Example 7**

Solutions: [Solutions](#)

Practice 4: Composition

Explain why multiplying a complex number by i rotates it by 90° CCW by writing i as a matrix. Then compute $(3 + 4i) \cdot i^3$ and describe the geometric transformation in words. What matrix corresponds to i^3 ?

→ Reference: **Example 1, 5**

Solutions: [Solutions](#)

Practice 5

Write $z = -1 + i\sqrt{3}$ in polar form. Compute z^6 . Write the matrix for z and for z^6 . What is $\det(\text{matrix for } z)$ and $\det(\text{matrix for } z^6)$? How are they related?

→ Reference: **Example 4, 6**

Solutions: [Solutions](#)

Practice 6

The three cube roots of -8 form a triangle in the complex plane. Find its area. Also find a 2×2 matrix that rotates the plane by 120° , and verify that applying it three times gives the identity. Connect this matrix to one of the cube roots.

→ Reference: **Example 7, 12A2 Example 13**

Solutions: [Solutions](#)

Practice 7: Matrix Bridge

The complex number $z = 3 + 4i$ corresponds to matrix $M = \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix}$. Find M^{-1} using the 2×2 inverse formula from 12A2. Then compute $1/z$ directly and verify the results match. What geometric operation does M^{-1} represent?

→ Reference: **Example 3, 8**

Solutions: [Solutions](#)

Practice 8: Real Battle

Prove that the product of all n th roots of unity (for $n \geq 2$) is $(-1)^{n-1}$. Test this for $n = 3, 4, 5$. Also: the n th roots of unity form a regular n -gon inscribed in the unit circle. Find a formula for its area in terms of n .

→ Reference: **Example 7**

Solutions: [Solutions](#)

Basic Algebra Drill — Complex Numbers (15 Problems)

Pure calculation + geometric insight. Build fluency with i , polar form, and the matrix connection.

- D1.** Simplify i^{27} . Write as 1 , -1 , i , or $-i$.
- D2.** Compute $(2 - 3i)(1 + 4i)$. Write in $a + bi$ form.
- D3.** Find the conjugate and modulus of $z = 5 - 12i$.
- D4.** Find i^{4k+3} for any integer k .
- D5.** Simplify $\frac{1}{i}$. Write in $a + bi$ form. What matrix does this correspond to?
- D6.** Compute $(3 + 2i) + (5 - 7i) - (1 + 4i)$.
- D7.** Find the modulus of $z = -6 + 8i$.
- D8.** Write $e^{i\pi/2}$, $e^{i\pi}$, $e^{i3\pi/2}$ in $a + bi$ form. What do you notice?
- D9.** Find $\bar{z}$ and $|z|$ for $z = -3i$.
- D10.** Simplify $i^{15} + i^{16} + i^{17} + i^{18}$. Explain why any four consecutive powers of i sum to 0.
- D11. ♦ Geometry** — Plot $z = 2 + 2i$ and its conjugate on the complex plane. What is the geometric relationship? What is $z \cdot \bar{z}$? What is $z + \bar{z}$?
- D12. ♦ Matrix Connection** — Write the matrix for $z = 1 + i$. Compute $\det(M)$. What is $|z|$? How are they related?
- D13. ♦ Geometry** — Multiply $z_1 = 1 + i$ by $z_2 = i$. What happens geometrically? Draw both z_1 and $z_1 \cdot z_2$ on the complex plane.
- D14. ♦ Matrix Connection** — The matrix for i is $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Compute J^2 , J^3 , J^4 . What complex numbers do these correspond to?
- D15. ♦ Geometry** — Find $1/z$ for $z = 2i$. Where is z relative to the unit circle? Where is $1/z$?

What is the product $|z| \cdot |1/z|$?

Solutions: [Solutions](#)

Advanced Algebra Drill — Complex Numbers (15 Problems)

Multi-step. Connect polar form, Euler, De Moivre, matrices, and geometry.

- A1.** Solve for z : $(1 + i)z + 3 = 2i - z$. Write z in $a + bi$ form. Write the corresponding matrix and compute its determinant.
- A2.** Compute $(1 + i\sqrt{3})^9$ using De Moivre. Give in $a + bi$ form. What is the corresponding matrix, and what rotation angle does it represent?
- A3.** Find all four 4th roots of -16 . Write in both $re^{i\theta}$ and $a + bi$ form. Write the matrix for one of the roots.
- A4.** The four 4th roots of 1 form a square. Compute its area. Compute the sum of all four roots.
- A5.** Write $z = \sqrt{3} - i$ in polar form. Compute z^6 . Write the matrix for z and confirm that its 6th power equals the matrix for z^6 .
- A6.** Solve $z^2 + 2z + 5 = 0$ over $\mathbb{C}$. Verify that the roots are conjugates.
- A7.** Find all complex z such that $z^3 = 8i$. Give in $a + bi$ form.
- A8.** If $z = 2e^{i\pi/6}$, find z^4 and $1/z$ in $a + bi$ form. Write the matrices and verify.
- A9.** Prove $|z_1 z_2| = |z_1| \cdot |z_2|$ using the matrix determinant: $\det(M_1 M_2) = \det(M_1) \det(M_2)$.
- A10.** The three cube roots of $8i$ form a triangle. Find its area.

◆ Geometry & Matrix Fusion (Problems A11–A15)

- A11. ◆** The complex number $z = \frac{1}{2} + \frac{\sqrt{3}}{2}i$ satisfies $z^3 = -1$ (check this). What rotation angle does z represent? Find a 2×2 matrix R such that $R^3 = -I$. What is $\det(R)$?

A12. ♦ Two complex numbers $z_1 = 3e^{i\pi/6}$ and $z_2 = 2e^{i\pi/3}$ are multiplied. Find $z_1 z_2$ in $a + bi$ form. Find the area of the triangle formed by 0, z_1 , and $z_1 z_2$ in the complex plane.

A13. ♦ Consider the set $\{z : |z| = 1\}$ (the unit circle). Show that this set is closed under multiplication: if $|z_1| = |z_2| = 1$, then $|z_1 z_2| = 1$. In matrix language: the set of pure rotation matrices is closed under multiplication — the product of two rotations is a rotation. Prove it.

A14. ♦ The n th roots of unity are $1, \omega, \omega^2, \dots, \omega^{n-1}$ where $\omega = e^{2\pi i/n}$. Show that $1 + \omega + \omega^2 + \dots + \omega^{n-1} = 0$ for $n \geq 2$. (Hint: geometric series formula.) Explain why this means the center of mass of the regular n -gon is at the origin.

A15. ♦ The transformation $T(z) = iz$ rotates every point by 90° . Write T as a 2×2 matrix acting on $\begin{pmatrix} x \\ y \end{pmatrix}$ where $z = x + iy$. Then consider $S(z) = \bar{z}$ (conjugation). Write S as a 2×2 matrix. Compute $T \circ S$ and $S \circ T$ as matrices. Do they commute? What is the geometric meaning of each composition?

Solutions: [Solutions](#)

Today's Procedure

Step 1: The matrix connection — $a+bi = [[a,-b],[b,a]]$.

$i = 90^\circ$ rotation matrix. $i^2 = -1 = 180^\circ$ rotation.

Every complex operation is a matrix operation you already know.

Step 2: Polar form — $z = re^{i\theta}$: $r = |z|$, $\theta = \arg(z)$.

Multiplication: multiply moduli, add angles.

De Moivre: $z^n = r^n e^{in\theta}$ (matrix power).

Step 3: Roots — $z^n = w$ has n solutions, evenly spaced on a circle.

Fundamental Theorem of Algebra: degree $n \rightarrow n$ complex roots.

Complex roots of real polynomials come in conjugate pairs.

How to Read These Symbols

Symbol	Reads as	Meaning
i	"i" / "the imaginary unit"	90° rotation matrix J ; $i^2 = -1$
$z = a + bi$	"z equals a plus b i"	shorthand for matrix $\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$
$\bar{z}$	"z bar" / "conjugate"	$a - bi$, reflection across real axis, matrix transpose
$ z $	"modulus of z"	$\sqrt{a^2 + b^2} = \sqrt{\det(M)}$, distance from origin
$\operatorname{Re}(z)$	"real part of z"	the a in $a + bi$
$\operatorname{Im}(z)$	"imaginary part of z"	the b in $a + bi$ (a real number, NOT bi !)
$re^{i\theta}$	"r e to the i theta" / "polar form"	stretch by r , rotate by θ
$\arg(z)$	"argument of z"	angle θ from positive real axis
Euler's formula	"Euler's formula"	$e^{i\theta} = \cos \theta + i \sin \theta = R_\theta$
De Moivre	"De Moivre's theorem"	$z^n = r^n e^{in\theta}$ — matrix power
n th roots of unity	"n-th roots of unity"	$e^{2\pi i k/n}$, $k = 0, \dots, n - 1$ — regular n -gon

Terminology

What we called it	Mathematical term	Notation
rotation-scaling matrix	complex number	$z = a + bi \leftrightarrow \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$

What we called it	Mathematical term	Notation
90° rotation	imaginary unit	$i, i^2 = -1$
transpose / reflection	complex conjugate	$\bar{z} = a - bi$
sqrt-of-determinant	modulus	$ z = \sqrt{a^2 + b^2}$
rotation angle	argument	$\theta = \arg(z)$
stretch + rotate form	polar / exponential form	$z = re^{i\theta}$
matrix power	De Moivre's theorem	$z^n = r^n e^{in\theta}$
regular polygon vertices	roots of unity	$e^{2\pi ik/n}$
